

HASTELLOY B-3 (UNS N10675) – Datasheet

Description:

Hastelloy B-3 is a high-stability nickel-molybdenum alloy designed for excellent resistance to strong reducing acids (hydrochloric, hydrobromic, sulfuric, phosphoric). It improves upon B-2 alloy with better thermal stability, resistance to pitting, stress corrosion cracking, and knife-line/HAZ attack after welding. Ideal for chemical processing, pickling, semiconductor, and pharmaceutical industries.

Chemical Composition

Element	Composition (%)
Nickel	65 (Balance)
Chromium	1.5
Molybdenum	27–32
Iron	1–3
Cobalt	3.0
Manganese	1.5
Silicon	0.10
Carbon	0.01
Phosphorus	0.2
Sulfur	0.01

Physical Properties

Property	Value
Density	9.22 g/cm ³
Melting Point	1370°C (2500°F)
Tensile Strength	110,000 psi / 760 MPa
Yield Strength	52,000 psi / 355 MPa
Elongation	40%

Mechanical Properties

Property	Metric	Imperial
Tensile Strength (@538°C)	760 MPa	110,000 psi
Yield Strength (@427°C)	355 MPa	52,000 psi
Elastic Modulus (RT)	210 GPa	30,000 ksi
Elongation at Break (@204°C)	40%	40%
Hardness, Rockwell B (plate)	88	88

Available Products

- Pipes: Seamless, ERW & EFW
- Tubes: Capillary, U-Bend
- Sheets & Plates
- Coils & Strips
- Bars: Round, Flat, Hex, Square
- Pipe Fittings: Elbows, Tees, Reducers, Caps, Stub Ends
- Flanges: Weld Neck, Slip-On, Blind, Socket Weld, Threaded
- Fasteners: Bolts, Nuts, Screws, Washers, Studs
- Custom Fabricated Components

Key Advantages

- Exceptional resistance to strong reducing acids
- Superior weldability – prevents knife-line attack and HAZ corrosion
- Maintains mechanical strength under high temperature and acidity
- Improved formability and fabrication over B-2
- Long service life in aggressive chemical environments

Applications

- **Chemical Processing:** Reactors, heat exchangers, evaporators, storage tanks
- **Pharmaceutical & Fine Chemicals:** Vessels, piping for ultra-pure environments
- **Metal Pickling & Acid Regeneration:** Acid recovery units, pickling lines
- **Electronics & Semiconductor:** Corrosive chemical baths, etching equipment
- **Agrochemical & Fertilizer Production:** Reactors, acid transfer lines
- **Pulp & Paper Industry:** Chemical pulping and bleaching stages